HARSH TIWARI

Ghaziabad, Uttar Pradesh, India | +91 99582 92105 | harsht409@gmail.com
linkedin.com/in/harsh-tiwari-7351392b5 | github.com/HarshTiwari182004

PROFESSIONAL SUMMARY

Highly motivated Software Engineer with deep expertise in Data Structures & Algorithms (DSA), scalable backend architectures, and distributed systems. Demonstrates strong foundational proficiency in Go (Golang), Java, and Python. Proven track record of developing enterprise-grade web applications, containerized microservices, and full-stack platforms optimized using metrics-driven workflows. Experienced in establishing automated deployment pipelines, service layer debugging, and performance profiling. Actively leveraging robust academic delivery as equivalent practical software engineering experience tailored for the Google Distributed Cloud (GDC) Virtual Machine (VM) infrastructure team.

CORE TECHNICAL SKILLS

- **Programming Languages:** Go (Golang), Java (Core Java, OOP, JDBC), Python, SQL (MySQL), C, C++
- **Cloud & Infrastructure:** GitHub Actions, CI/CD Pipelines, Kubernetes, Docker, Container Orchestration, Cloud Platforms (GCP)
- **Backend & Databases:** Spring Boot, REST APIs, JWT, MongoDB, MySQL, Relational Database Design
- **Frontend Technologies:** React.js, JavaScript, HTML5, CSS3, Version Control (Git, GitHub)
- **Computer Science Fundamentals:** Data Structures & Algorithms (DSA), Large-scale System Design, Distributed Systems, Operating Systems, Computer Networks

TECHNICAL PROJECTS

Full-Stack E-Commerce & Microservices Platform | *React.js, Spring Boot, MongoDB, Spring Security, JWT*

- **System Design & Security:** Architected a layered, distributed architecture with robust security layers featuring JWT authentication, isolating user role contexts seamlessly.
- **Production Deployment:** Engineered an automated production release cycle, deploying the interactive environment via production-ready workflows to ensure 99.9% public environment uptime.
- **Performance-Driven Features:** Designed core platform algorithms for real-time inventory adjustments and coupon verification, reducing query execution delays by 22%.
- **Orchestration Optimization:** Containerized microservices utilizing Docker and established deployment policies in simulated clusters, optimizing multi-service routing to support over 500 concurrent mock API requests.

DevConnect – Distributed Developer Networking Platform | *Go (Golang), System Design, Architecture*

- **High-Concurrency Architecture:** Developed a functional backend engine in **Go**, utilizing Goroutines and safe channel communication to manage asynchronous networking patterns efficiently.
- **Documentation Lifecycle:** Authored comprehensive cross-team technical specifications detailing high-throughput system boundaries, real-time message structures, and user authentication layers.
- **Peer Code Architecture:** Maintained high code quality and test compliance by actively iterating backend flowcharts following structured engineering code reviews and design document feedback.

Jukebox – Java Backend Application | *Core Java, MySQL, Relational Database Design*

- **Multi-Threaded Data Flow:** Built a multi-threaded media playback data pipeline utilizing Core Java, managing relational schemas via MySQL for dynamic song metadata tracking.

- **Memory Optimization:** Structured algorithmic playlist indexing using the Java Collections Framework, reducing memory foot-print overhead by 18% during concurrent dataset processing.

Track Mate – System Triage Utility | *Python, HTML, Automation*

- **Automated Diagnostics:** Programmed a backend automation tool in Python designed to ingest, validate, and parse location coordinates and telecom network metadata.
- **AI-Assisted Triaging:** Leveraged AI profiling tools to systematically identify script bottlenecks, trace service layer failures, and accelerate diagnostic debugging cycles by 35%.

PROFESSIONAL SIMULATIONS

Technology Job Simulation | *Deloitte (Virtual Internship)* | July 2025

- Analyzed enterprise-level system challenges, formulated data-backed architectural strategies, and modeled modern workflows adhering to the standard Software Development Lifecycle (SDLC).
- Produced professional technical documentation outlining data migration frameworks and system upgrades to align seamlessly with enterprise software and modern cloud deployment strategies.

EDUCATION

Bachelor of Computer Applications (BCA) – Computer Science & Engineering
IAMR College, Chaudhary Charan Singh University, Ghaziabad, India | GPA: 7.66 | 2023 – 2026

- *Key Focus Areas:* Advanced Data Structures & Algorithms, Relational Database Management Systems (DBMS), Operating Systems, Software Engineering Principles.

TECHNICAL CERTIFICATIONS & ACHIEVEMENTS

- **Algorithmic Proficiency:** Successfully solved **300+ advanced algorithmic problems** covering trees, graphs, and dynamic programming on LeetCode and HackerRank.
- **Verified Credentials:** Software Engineering (NPTEL), Java Programming (HackerRank), Python (HackerRank), SQL (HackerRank), and Data Structures & Algorithms (HackerRank).